

CURRICULUM VITAE

ANKUR SAHARIA (b. 1985)

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Education

Doctorate, PhD in Integrated Optics (ECE) from MNIT Jaipur Jaipur, 2021

Major: Integrated Optics

Research Interests: Optical Communication Engineering: All Optical Switches, Optical Logic Gates & Circuits, Photonic Crystal Fibers, Nanophotonics, Nano-optical structures, Sensors, Integrated Photonics, Structured light, OAM generation and propagation, AI for Photonics.

M.Tech, ECE MNIT Jaipur, India Jaipur, 2014

Major: ECE

Research Interests: Optical Filters, Optical amplifier, Optical logic circuit;

Bachelor in Engineering (B.E.), University of Rajasthan, Jaipur, India

Jaipur, 2008

Major: Electronics & Telecommunications

Research Interests

Optics/Photonics: Optoelectronics, Optical waveguides, Photonic crystal Fibers, Optical sensors, Bio-photonics, Quantum Optics.

Professional Experience and Employment

April 2021- present	Assistant Professor	Manipal University, Jaipur-campus, India
July 2017- Jan 2021	Teaching Assistant (PhD Student)	MNIT Jaipur, India
July 2014 – July 2017	Assistant Professor (Senior Scale) ,	Poornima Institute of Engineering & Technology Jaipur, India
July 2011 – 2014	Post Graduate Student	MNIT Jaipur, India
July 2009-2011	Lecturer, Department of ECE, Poornima Institute of Engineering & Technology Jaipur, INDIA	Poornima Institute of Engineering & Technology Jaipur, India
July 2004 to July 2008	Graduate Student	University of Rajasthan, Jaipur, India

Information about Employers

Manipal University, Jaipur-campus, India	https://jaipur.manipal.edu/	Academic & Research Institute
MNIT Jaipur, India	https://mnit.ac.in/	Academic & Research Institute
Poornima Institute of Engineering & Technology Jaipur, India	http://www.piet.poornima.org/	Academic Institute

Special Skills / Professional Interests

Equipment: fiber optic splice and test systems; fusion splicers, optical time domain reflectometers, optical power meters, chromatic dispersion analyzer, etc.

Publications

Papers: Nearly 50 printed papers in technical journals and proceedings; 2 book chapters; *h-index*: Google scholar – 8; Scopus – 8.

Thesis: Ankur Saharia, *Microring resonator-based structuring to implement optical digital components and circuits for Inchip applications*- 2021(Ph.D)
Ankur Saharia, *Realization of Optical Logic gate using semiconductor optical amplifier and bandpass filter* (M.Tech)-2014

Grants and Awards

S.N	Title	Cost in Lakh	Month of submission	Role as PI/Co-PI	Agency	Status
1.	OAM INDUCED QUANTUM TELEPORTATION	Rs. 675400.00	Sept 2022	PI	DST-JSPS	Accepted for Evaluation
2.	Microring resonator integrated optical quantum states generation.	Rs. 26,25,240	April 2022	PI	SERB	Not Selected
3.	Development Of Information Processing and ultrafast decision making circuits Using Quantum Optics	Rs.20,55,240	Dec 2021	PI	ISRO	Qualified Screening
4.	TERAHERTZ FREE SPACE COMMUNICATION LINK FOR QUANTUM NETWORKS	Rs. 628100	Sept 2021	PI	DST-JSPS	Not selected
5.	Travel Grant		2023	-	Optica	Awarded
6.	Travel Grant- Fio-LS		2019	-	Optica	Awarded

Committee Member Position

1. Convenor International Conference “International Conference on Recent Cognizance in Wireless Communication and Image Processing”-2014 ,Jan 16-17,2015.
2. Convener National Conferences, Convergence- 2016, Convergence- 2017
3. Coordinator IEEE conference MITE-2013 ,Dec 22-23,2014.
4. Committee Member- OWT-2020, OWT-2019
5. Reviewer of IEEE Access, Reviewer of IEEE Sensor Journal, Reviewer of IEEE Transaction of Circuits & Systems, Reviewer of Optik Journal- Elsevier, Reviewer of Microsystems Technologies , Reviewer of Results in Optics, Reviewer of Optical Engineering, Reviewer of Biomedical Optics, Reviewer of Optical and Quantum Electronics, Reviewer of OAM-RC journal, Reviewer Optica Continuum
6. Reviewer Board member Frontier in Physics

Professional Societies

2017 – present The Optical Society (OSA) – Senior Member
2018 – present The Society of Photo-Optical Instrumentation Engineers (SPIE)
2018-2022 Member- IEEE and IEEE Photonics society

List of Patents granted (National and International). : 1 –(International)

1. 2021104782: **Internet of Things (IoT) sensors-based system for child monitoring & method thereof**

Main Publications (the Last 5 years)

1. Ankur Saharia, Nitesh Mudgal, Kamal K Choure, Ankit Agarwal, Jalil Ali, Preecha P. Yupapin, Ravi Kumar Maddila, Ghanshyam Singh "Elementary magnitude comparator and flip-flop using Si₃N₄ based Microring resonator" *Optoelectronics & Advance Materials-Rapid Communications* , Vol. 14, No. 1-2, pp. 43-52, Jan.-Feb. 2020.
2. Ankur Saharia, Manish Tiwari, Ashish Kumar Ghunawat, Anton Bourdine, Vladimir Burdin, Ravi Kumar Maddila, Ghanshyam Singh "Evolution of adder and subtractor circuit using Si₃N₄ Microring Resonator" *Journal of Optical Communication* , <https://doi.org/10.1515/joc-2019-0028>, October 19, 2019.
3. Ankur Saharia, Jalil Ali, Preecha P. Yupapin, Ravi Kumar Maddila Ghanshyam Singh "An Elementary Optical Logic Circuit for Quantum computing: A Review" *Optical and Quantum Electronics* , 51:224, pp. 1-13, <https://doi.org/10.1007/s11082-019-1944-3>, 2019.
4. Ankur Saharia, Nitesh Mudgal, Ravi kumar Maddila Ghanshyam Singh Elementary reflected code converter using a silicon nitride-based microring resonator. *J Comput Electron* (2021). <https://doi.org/10.1007/s10825-020-01644-w>.
5. Ankur Saharia, Nitesh Mudgal, Kamal Kishor Choure, Ravikumar Maddila, Manish Tiwari, Ghanshyam Singh, "Proposed all-optical read-only memory element employing Si₃N₄ based optical microring resonator," *Optik*, Volume 251, 2022, 168493, ISSN 0030-4026, doi.org/10.1016/j.ijleo.2021.168493.
6. Kamal Kishor Choure, Ankur Saharia, Nitesh Mudgal, Rahul Pandey, Ankit Agarwal, Manisha Prajapat, Ravikumar Maddila, Manish Tiwari, Ghanshyam Singh, Reconfigurable and compact reversible channel multiplexers using Si₃N₄ based optical microring resonator, *Optics Communications*, 2022, 129126, , <https://doi.org/10.1016/j.optcom.2022.129126>.
7. Choure, K.K., Bharti, G.K., Saharia, A. et al. Design and simulation of all-optical Swap and Fredkin gates using mode-rotation based race-track ring resonator. *Opt Quant Electron* 54, 276 (2022). <https://doi.org/10.1007/s11082-022-03662-3>.
8. N. MUDGAL, K. K. CHOURE, A. SAHARIA, A. AGARWAL, M. K. FALASWAL, S. SAHU, S. V. SINGH, G. SINGH, Comparative assessment of fiber SPR sensor for sensitivity enhancement using BaTiO₃ layer, *Optoelectronics and Advanced Materials - Rapid Communications*, 16, 3-4, March-April 2022, pp.114-120 (2022).
9. Mudgal, N., Choure, K.K., Falaswal, M.K. et al. Impact of Taguchi Optimization in Fiber Surface Plasmon Resonance Sensors Based on Si₃N₄ Layer. *Braz J Phys* 52, 80 (2022). <https://doi.org/10.1007/s13538-022-01088-6>.
10. Mudgal, N., Saharia, A., Agarwal, A. et al. Modeling of highly sensitive surface plasmon resonance (SPR) sensor for urine glucose detection. *Opt Quant Electron* 52, 307 (2020). <https://doi.org/10.1007/s11082-020-02427-0>
11. N. Mudgal, Ankur Saharia, Kamal Kishor Choure, Ankit Agarwal, G. Singh, "Sensitivity enhancement with anti-reflection coating of Silicon nitride (Si₃N₄) layer in Silver based Surface Plasmon Resonance (SPR) sensor for sensing of DNA hybridization", *Applied Physics A* 126(12), DOI: 10.1007/s00339-020-04126-9, December 2020.
12. Nitesh Mudgal, Ankur Saharia, Ankit Agarwal, Ghanshyam Singh, "Zno and bi-metallic(ag-Au) layers based surface plasmon resonance biosensor with BaTiO₃ and graphene for biosensing application", *IETE Journal of Research* <https://doi.org/10.1080/03772063.2020.1844074>- 2019.
13. Ankur Saharia, Choure, K., Mudgal, N. et al. Introductory Review on All-Optical Machine Learning Leap in Photonic Integrated Circuits. *Opt. Mem. Neural Networks* 31, 393–402 (2022). <https://doi.org/10.3103/S1060992X22040075>
14. Amit Kumar Singh, Shweta Mittal, Mangal Das, Ankur Saharia, Manish Tiwari, Optical biosensors: a decade in review, *Alexandria Engineering Journal*, Volume 67, 2023, Pages 673-691, ISSN 1110-0168, <https://doi.org/10.1016/j.aej.2022.12.040>.
15. Mittal, S.; Saharia, A.; Ismail, Y.; Petruccione, F.; Bourdine, A.V.; Morozov, O.G.; Demidov, V.V.; Yin, J.; Singh, G.; Tiwari, M. Spiral Shaped Photonic Crystal Fiber-Based Surface Plasmon Resonance Biosensor for Cancer Cell Detection. *Photonics* 2023, 10, 230. <https://doi.org/10.3390/photonics10030230>.
16. Bourdine, A.V.; Demidov, V.V.; Ter-Nersesyants, E.V.; Pchelkin, G.A.; Shurupov, D.N.; Khokhlov, A.V.; Matrosova, A.S.; Kashin, A.I.; Bureev, S.V.; Dashkov, M.V.; Evtushenko, A.S.; Zaitseva, E.S.; Gizatulin, A.R.; Meshkov, I.K.; Dyavangoudar, A.A.; Saharia, A.; Tiwari, M.; Vasilets, A.A.; Elagin, V.S.; Singh, G.; Dukelskii, K.V. Twisted Silica Few-Mode Hollow GeO₂-Doped Ring Core Microstructured Optical Fiber. *Preprints.org* 2023, 2023051814. <https://doi.org/10.20944/preprints202305.1814.v1>.
17. Amogh A Dyavangoudar, Mayur Kumar Chhipa, Ankur Saharia, Yaseera Ismail, Francesco Petruccione, Anton V. Bourdine, Oleg G. Morozov, Vladimir V. Demidov, Juan Yin, Ghanshyam Singh and Manish Tiwari "Orbital Angular Momentum Mode Propagation and Supercontinuum Generation in a Soft Glass Bragg Fiber" Accepted, *IEEE Access*, April 2023.

18. Parashar, D. Yadav, A. Saharia, M. Tiwari and G. Singh, "Implementation of four-band slotted patch Antenna," 2022 14th International Conference on Computational Intelligence and Communication Networks (CICN), Al-Khobar, Saudi Arabia, 2022, pp. 1-4, doi: 10.1109/CICN56167.2022.10008260.
19. A. Dyavangoudar, A. Sharma, A. Saharia, Y. Ismail, A. V. Bourdine and M. Tiwari, "Orbital Angular Momentum Mode Propagation in a Hexagonal-Ring Core Spiral PCF," 2022 6th International Conference on Electronics, Communication and Aerospace Technology, Coimbatore, India, 2022, pp. 189-193, doi: 10.1109/ICECA55336.2022.10009239.
20. T. Sharma et al., "A Photonic Crystal Fiber for Chemical Sensing Applications," 2022 4th Novel Intelligent and Leading Emerging Sciences Conference (NILES), Giza, Egypt, 2022, pp. 122-125, doi: 10.1109/NILES56402.2022.9942442.
21. A. A. Dyavangoudar, S. Punia, A. V. Bourdine, Y. Ismail, G. Singh, A. Saharia, and M. Tiwari, "A Snowflake shaped Photonic Crystal Fiber for OAM mode propagation," in *Frontiers in Optics + Laser Science 2022 (FIO, LS)*, Technical Digest Series (Optica Publishing Group, 2022), paper JW5A.46.
22. S. Mittal, P. Shrivastava, A. Saharia, and M. Tiwari, "Hexagonal Spiral shaped PCF based on Surface Plasmon Resonance biosensor," in *Frontiers in Optics + Laser Science 2022 (FIO, LS)*, Technical Digest Series (Optica Publishing Group, 2022), paper JW5A.67
23. Mali, R. et al. (2023). Emergence of Silicon Photonics in the Field of Biomedical Sensing Applications: A Review. In: Dwivedi, S., Singh, S., Tiwari, M., Shrivastava, A. (eds) *Flexible Electronics for Electric Vehicles*. Lecture Notes in Electrical Engineering, vol 863. Springer, Singapore. https://doi.org/10.1007/978-981-19-0588-9_33
24. Pandey, R. et al. (2023). A High Bit Rate Half Adder Based on Square Lattice Photonic Crystal. In: Tiwari, M., Ismail, Y., Verma, K., Garg, A.K. (eds) *Optical and Wireless Technologies. OWT 2021*. Lecture Notes in Electrical Engineering, vol 892. Springer, Singapore. https://doi.org/10.1007/978-981-19-1645-8_38
25. Choure, K.K., Bharti, G.K., Saharia, A., Mudgal, N., Singh, G. (2023). Modelling of SWAP Gate Using Compact Ring Resonator. In: Tiwari, M., Ismail, Y., Verma, K., Garg, A.K. (eds) *Optical and Wireless Technologies. OWT 2021*. Lecture Notes in Electrical Engineering, vol 892. Springer, Singapore. https://doi.org/10.1007/978-981-19-1645-8_39
26. garwal, A. et al. (2023). Designing of GaN Based Photonic Crystal Biosensor for Blood Disease Analysis. In: Tiwari, M., Ismail, Y., Verma, K., Garg, A.K. (eds) *Optical and Wireless Technologies. OWT 2021*. Lecture Notes in Electrical Engineering, vol 892. Springer, Singapore. https://doi.org/10.1007/978-981-19-1645-8_28
27. S. Mittal, P. Shrivastava, A. Tummuri, A. Saharia, and M. Tiwari, "A Surface Plasmon Resonance drew Biosensor based on U-Shape Photonic Crystal Fiber," in *Imaging and Applied Optics Congress 2022 (3D, AOA, COSI, ISA, pcAOP)*, Technical Digest Series (Optica Publishing Group, 2022), paper JW2A.8.
28. Ashish Tummuri, Shweta Mittal, Ankur Saharia, Manish Tiwari, "Numerical analysis of octagonal photonic crystal fiber for orbital angular momentum propagation," *Proc. SPIE 12140, Micro-Structured and Specialty Optical Fibres VII*, 121400G (19 May 2022); <https://doi.org/10.1117/12.2621422>
29. P. Shrivastava et al., "Photonic Crystal Fiber-based Temperature Sensor," 2022 8th International Conference on Advanced Computing and Communication Systems (ICACCS), 2022, pp. 871-874, doi: 10.1109/ICACCS54159.2022.9784995.
30. A. Aggarwal, R. Gupta, A. Saharia and M. Tiwari, "Hexagonal Spiral Photonic Crystal Fiber," 2021 International Conference in Advances in Power, Signal, and Information Technology (APSIT), 2021, pp. 1-4, doi: 10.1109/APSIT52773.2021.9641277.
31. Ankur Saharia, Kamal Kishor Choure, Nitesh Mudgal, Rahul Pandey, Dinesh Bhatia, Manish Tiwari, Ghanshyam Singh, "Integrated Photonics for Artificial Intelligence Applications" *VLSI and Hardware Implementations Using Modern Machine Learning Methods*, 293-303, CRC Press-2022.
32. Choure K.K., Saharia A., Mudgal N., Tiwari M., Singh G. (2022) Recent Advancement in High Speed and Secure Quantum Key Distribution: A Review. In: Tiwari M., Maddila R.K., Garg A.K., Kumar A., Yupapin P. (eds) *Optical and Wireless Technologies. Lecture Notes in Electrical Engineering*, vol 771. Springer, Singapore. https://doi.org/10.1007/978-981-16-2818-4_29
33. Trivedi D., Saharia A., Choure K., Tiwari M., Maddila R.K., Singh G. (2022) Three-Qubit Implementation of Quantum Fourier Transform for Shor's Algorithm. In: Tiwari M., Maddila R.K., Garg A.K., Kumar A., Yupapin P. (eds) *Optical and Wireless Technologies. Lecture Notes in Electrical Engineering*, vol 771. Springer, Singapore. https://doi.org/10.1007/978-981-16-2818-4_12.
34. Jangid M., Saharia A., Nitesh Mudgal, Singh S.V., Singh G. (2022) Performance Enhancement of Surface Plasmon Resonance (SPR) Structure Using a Sinusoidal Diffraction Grating. In: Tiwari M., Maddila R.K., Garg A.K., Kumar

- A., Yupapin P. (eds) Optical and Wireless Technologies. Lecture Notes in Electrical Engineering, vol 771. Springer, Singapore. https://doi.org/10.1007/978-981-16-2818-4_13
35. Saharia A., Maddila R.K., Ismail T., Fahim I.S., Tiwari M., Singh G. (2020) "Parametric analysis of optical microring resonator" In: Janyani V., Singh G., Tiwari M., Ismail T. (eds) Optical and Wireless Technologies. Lecture Notes in Electrical Engineering, vol. 648, pp. 307-315, https://doi.org/10.1007/978-981-15-2926-9_34, Springer Singapore, May 2020 (SCOPUS INDEX).
 36. Saharia, R. K. Maddila, and G. Singh, "Elementary novel all-optical reversible logic gate using Si₃N₄ based optical microring resonator," in *Frontiers in Optics + Laser Science APS/DLS, OSA Technical Digest (Optical Society of America, 2019)*, paper JT_u3A.126, <https://doi.org/10.1364/FIO.2019.JTu3A.126>, September 2019 (SCOPUS INDEX).
 37. Ankur Saharia, Ashish Kumar Ghunawat, Anton V. Bourdine, Vladimir A. Burdin, Manish Tiwari, Ravi Maddila, Ghanshyam Singh, "Cascading application of all optical Universal NAND gate for implementation of basic logic gates", 3rd IEEE Conference ICRAIE-2018, Poonima University Jaipur, IEEE Xplore Digital Library, <https://doi.org/10.1109/ICRAIE.2018.8710442>, 2019.
 38. Saharia A., Singh G., et al. (2020) A Comparative Study of Various All-Optical Logic Gates. In: Janyani V., Singh G., Tiwari M., d'Alessandro A. (eds) Optical and Wireless Technologies. Lecture Notes in Electrical Engineering, vol 546, pp 429-437, https://doi.org/10.1007/978-981-13-6159-3_45, Print ISBN: 978-981-13-6158-6, Electronic ISBN: 978-981-13-6159-3, Springer, Singapore, April 10, 2019 (SCOPUS INDEX)..
 39. N. Mudgal , A.Saharia ,A.Agarwal, S.Sahu , A.K Ghunawat., G Singh. "Comparative Study of Interferometer and Ring Resonator Based Biosensors": A Review. In: Janyani V., Singh G., Tiwari M., d'Alessandro A. (eds) Optical and Wireless Technologies. Lecture Notes in Electrical Engineering, vol 546, pp 419-427, https://doi.org/10.1007/978-981-13-6159-3_44, Print ISBN: 978-981-13-6158-6, Electronic ISBN: 978-981-13-6159-3, Springer, Singapore, (2018).
 40. Prajapat M., Saharia A., Singh G. (2020) Design of Half Adder Using Electro-Optic Effect in Mach–Zehnder Interferometer. In: Janyani V., Singh G., Tiwari M., Ismail T. (eds) Optical and Wireless Technologies. Lecture Notes in Electrical Engineering, vol 648. Springer, Singapore. https://doi.org/10.1007/978-981-15-2926-9_29
 41. Saini J.K., Saharia A., Ismail T., Fahim I.S., Tiwari M., Singh G. (2020) Polarization Encoded Multi-logic Functions with Direct Detection. In: Janyani V., Singh G., Tiwari M., Ismail T. (eds) Optical and Wireless Technologies. Lecture Notes in Electrical Engineering, vol 648. Springer, Singapore. https://doi.org/10.1007/978-981-15-2926-9_33